

- 11 A speed boat has two identical motors. When both motors are working, the speed boat attained a maximum speed of 36.0 m s^{-1} . Given that the drag force on the speed boat is proportional to the square of the speed, what is the maximum speed of the boat when only one motor is working.

A 9.0 m s^{-1}

B 18.0 m s^{-1}

C 24.2 m s^{-1}

D 28.6 m s^{-1}

Ans: D

Using $P = F_{\text{driving}} v$

At max speed, $F_{\text{driving}} = \text{Drag Force} = kv^2$,

Thus $P = (kv^2)v = kv^3$

$$\frac{P_1}{P_2} = \frac{v_1^3}{v_2^3}$$

$$\sqrt[3]{\frac{1}{2}} = \frac{v_1}{36} \text{ , } v_1 = 28.6 \text{ m s}^{-1}$$

